Shruti Garg

sgrg@alum.mit.edu • (+1) 732-858-4734
www.linkedin.com/in/sgrg914
Languages: Hindi + English (native), Mandarin (conversational)

Education

Massachusetts Institute of Technology (MIT) — **Cambridge, MA**
M.Eng + B.Sc Electrical Engg and CS: Robotic Manipulation, Underactuated Robotics, Multiagent Learning, Matrix Methods, Visual Navigation for Autonomous Systems, Microcontroller Project Lab
GPA: 4.8/5.0
2025

Edison High School Academy *Valedictorian, Scholar @ Columbia University Science Honors Program , Research Scholar at NJ Governor's School of Engg and Tech. (2019) @ Rutgers University* — **NJ,** 2020

Relevant Technical Skills

Python, Rust, SystemVerilog, **C++ | Robotics Algorithms for Planning & Control | OpenCV, ROS, drake toolbox**

Technical Experience

NVIDIA — *Robotics Systems Software Engineer* — **Santa Clara, CA**
Jun 24 - Sept 24, Sep 25 - Sep 26
bash scripting, systems work ROS, c++

Debugging issues on hardware and in simulation blocking deployment

- Inference optimization for GR00T / Cosmos models
- Pre-release benchmarking infra for Isaac ROS
- Deploying GR00T foundation model on Edge compute aka Thor
- System Monitors - Proof of Concept for the Greenwave Monitors

IISc ARTPARK — *MIT MISTI Intern* — **Bengaluru, India**
July- Aug '25

Worked with Prof. Amrutur Bharadwaj. Spent 2 months at the Startup incubator out of India's premier research institution. Visited a number of customers who wanted lab automation and understood their needs + constraints. Interfaced with startup for teleoperation and data collection

MIT Robot Locomotion Group CSAIL — **Cambridge, MA**
Sept23 - Oct25
Drake, Python+some jax, Optimization and Motion Planning

Mentor: Thomas Cohn, PI: Prof. Russell Tedrake, thesis link https://dspace.mit.edu/handle/1721.1/164827
Optimizing Manipulation Trajectories under non-isometric parameterizations + Motion Planning for Constrained Bimanual Manipulation. **IK for Teleoperation of non-holonomic manipulators**
First author journal paper published in RAL and presented in IROS Hangzhou 2025.
Funded as a MIT EECS | Lincoln Laboratory Undergraduate Research and Innovation Scholar

Anduril Industries — *Robotics Software Intern* — **Costa Mesa, CA**
Summer 2023
Rust, NixOS, systems work

gimbal simulator
tracer bullet approach: extending a **rust daemon** across interfaces/product different stacks to deliver a smooth synthetic video feed. (Did not know about rust before starting this project!)
remote access to hardware testing stack for Roadrunner
a NUC NixOS remote host for to access the mission PC and control the power supply

Honest AI (ML real estate startup) — *SWE and Machine Learning Intern* — **London, UK**
Summer 2022

ETL pipeline to prepare London Housing Data for a Language model powered real estate search tool. Communicated with external data providers to acquire an appropriately extended data set.

Arcturus Robotics- MIT's Autonomous Boat Team — *Electrical Engineering Lead* — **Fall 2022 - Summer 2023**
Hardware debugging and teamwork :)

Njord Competition, Trondheim Norway
In 2 months between competitions : overhauled our boat's electrical design, organized team for tight deadlines for (electrically) building multiple new boats, designed and integrated subsystems a MOSFET based signaling tower controller to indicate drive modes.

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Jointly responsible for all electrical systems on board our boat for Roboboat 2023 in Florida. Team **won best technical design** along with other awards.

Teaching Assistant @ MIT **2024**

6.4210 Robotic Manipulation: Perception Planning and Control Teaching Assistant
Running OH, grading, advising student projects, revising and prepping problem sets

6.310 Dynamical System Modeling and Control Design Lab Assistant
Staffed OH and Lab sessions, answering questions and helping students with labs and class concepts

Viziverse (early Augmented Reality startup) *Pose detection & tracking* **Boston, Spring 2022**
Worked with the founder to deliver a self-contained **first stage prototype from ground up for pitching to Angel investors**. OpenCV, python, integrating ML models.

NJ Governor's School of Engineering and Technology @ Rutgers University, research presented at **IEEE MIT Undergraduate Research Technology Conference 2019.** **NJ** 2019

Leadership, People Skills and Teaching

Student Government in East Campus (residential dormitory @ MIT) **2021-2022**

- ➔ *Executive Board (Treasurer)* Oversaw a 60k+ budget used by many committees and events. Advocated for residents and negotiated to achieve smooth operation of a building of 300+ people and an active counterculture hub. Chaired multiple other smaller task groups and committees.
- ➔ Organized MIT's Pumpkin Drop 2021 oversaw operations and approvals for dropping 100 pumpkins off the top of Tang Hall.
- ➔ *Hall Chair, Resident Peer Mentor* for my floor. Major point of communication between students and administration.
- ➔ *REX Chair* managed safety approvals and execution of student designed and lead builds including a rollercoaster.

Other Projects

Bimanual Rubik's Cube Solver Low level planning and control for an iiwa in simulation. **2022 Fall**
Presentation video here (code notebook in the description).

Underactuated Staff Spinner Optimizing for a controller to throw and catch a staff in a loop. Presentation video here (code notebook in the description) **Spring 2023**

Action controlled theater systems hand motion spinning a platform and panning sound. I2C, serial comms, speakers and motors among other things Proposal here **2022 Spring**